Dual-Frequency Autoencoder for Anomaly Detection in Transformed Hyperspectral Imagery

Yidan Liu, Weiying Xie, *Member, IEEE*, Yunsong Li, *Member, IEEE*, Zan Li, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Hyperspectral anomaly detection (HAD) is a challenging task since samples are unavailable for training. Although unsupervised learning methods have been developed, they often train the model using an original hyperspectral image (HSI) and require retraining on different HSIs, which may limit the feasibility of HAD methods in practical applications. To tackle this problem, we propose a dual-frequency autoencoder (DFAE) detection model in which the original HSI is transformed into high-frequency components (HFCs) and low-frequency components (LFCs) before detection. A novel spectral rectification is first proposed to alleviate the spectral variation problem and generate the LFCs of HSI. Meanwhile, the HFCs are extracted by the Laplacian operator. Subsequently, the proposed DFAE model is learned to detect anomalies from the LFCs and HFCs in parallel. Finally, the learned model is well-generalized for anomaly detection from other hyperspectral datasets. While breaking the dilemma of limited generalization in the sample-free HAD task, the proposed DFAE can enhance the background–anomaly separability, providing a better performance gain. Experiments on real datasets demonstrate that the DFAE method exhibits competitive performance compared with other advanced HAD methods.**

***Index Terms*— Autoencoder (AE), dual-frequency, hyperspectral anomaly detection (HAD), transformation.**

I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D data cube in which two dimensions show spatial information of remotely sensed scenes, and the third dimension provides spectral information containing the essential attributes of different materials, with hundreds of contiguous spectral bands. With high spectral resolution, hyperspectral imaging is widely applied in civilian and military fields such as precision agriculture [1], medical diagnosis [2], resource exploration [3], and land classification and management [4]. Depending on application requirements, diverse data analysis methods are derived, including scene classification [5], spectral unmixing [6], and target detection [7]. Among them, hyperspectral anomaly detection (HAD), as a special target detection method, can probe targets of interest that are significantly different from the surrounding background in both the spectral and spatial domains without any prior knowledge, which has received increasing attention from researchers [8].

The existing HAD methods can be classified into two major categories: traditional and deep-learning-based methods. The most classical traditional anomaly detection algorithm is the RX detector [9] which assumes that the background follows the multivariate Gaussian distribution. Based on this assumption, the RX detector first estimates the mean and variance of the entire image to establish the background model, and then calculates the Mahalanobis distance between the test pixel and the background to determine anomalies. Compared with the RX detector, the kernel RX (KRX) algorithm [10] represents the distribution of background data with a much more complex model by transforming each spectral pixel into a high-dimensional feature space. Although the KRX detector yields superior performance, it is highly computationally demanding. The cluster KRX (CKRX) [11] is a fast version of KRX, which replaces each pixel with its cluster's center to simplify computation. Not satisfied with the contamination of the background statistics by anomaly pixels in the above detectors, the local RX detector (LRX) [12] and the collaborative representation detector (CRD) [13] have been proposed to model the background by adopting a dual-window strategy. In addition, the low-rank and sparse matrix decomposition model (LSDM) has been extensively studied in HAD [14]–[17]. In [14], the robust principal component analysis (RPCA) [18] is used to decompose the HSI, and the RX detector is applied to detect anomalies on the sparse component. As a generalization of RPCA, the low-rank representation (LRR) model can recover more complex scenes in multiple subspaces by choosing an appropriate dictionary [15]. Instead of using the raw data directly for anomaly detection, [16] applies spectral unmixing algorithms to obtain abundance vectors as features of HSI to construct the background dictionary. Recently, Li *et al.* [17] modified LSDM with a Gaussian mixture model, which can describe noise characteristics more accurately, thereby separating anomalies apart from noise. Considering that HSIs also contain rich spatial information, both attribute and edge-preserving filtering-based anomaly detection (AED) [19] and structure tensor and guided filter (STGF) [20] algorithms use an attribute filter to detect anomalies with a specific area in the spatial domain. More recently, a prior-based tensor approximation (PTA) method has been proposed, which uses a third-order tensor to preserve the data structure for integrated

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consideration of all the dimensions [21]. Different from the above works, some interesting studies have also emerged in recent years. The fractional Fourier entropy (FrFE)-based HAD method [22] detects anomalies in an intermediate domain between the original reflectivity spectrum and its Fourier transform with the FrFE theory. Chang *et al.* [23] proposed an isolation-based discriminative forest model with subsampling to avoid background contamination caused by abnormal targets. In [24], kernel theory and union dictionary representation are combined to achieve anomaly detection, fully accounting for the intrinsic nonlinear characteristics of HSIs. In summary, determining the difference between anomaly and background through effective feature representation is crucial for HAD.

The deep-learning-based methods can capture nonlinear and latent features and estimate complex functions through hierarchical structures, which have become increasingly popular in anomaly detection [25]–[27]. Due to insufficient samples and inaccurate labeling, unsupervised learning models, like autoencoder (AE) and its various extended versions, have been widely used to achieve the HAD task. For example, Xie *et al.* [28] proposed a spectral constrained adversarial AE [SC_adversarial autoencoder (AAE)] to obtain hidden layer features with better discrimination while building a bilayer architecture to suppress the background. Considering the locality property of HSI, Lu *et al.* [29] embedded a learned manifold in the AE network and then calculated the reconstruction errors to detect anomalies. Wang *et al.* [30] also used reconstruction errors as anomalies and designed a fully convolutional AE with skip connections to achieve autonomous HAD. Besides, the spectral angle distance is taken as an initial anomaly score in [31], which can preferably reflect the spectral similarity of each pixel between the input and the output of variational AE. With the development of generative adversarial networks (GANs), this promising method of approximating complex distribution is used in HAD for background modeling [32], [33]. The above-mentioned deep-learning-based methods can capture data characteristics better than the traditional methods. However, most unsupervised learning methods need to retrain the model on different HSIs and adjust parameters for specific datasets, which means they can only detect anomalies on the current input HSI. The training process and parameter tuning will take a long time and consume lots of resources, making the feasibility of such HAD methods limited in practical applications. Therefore, it is highly essential to design a new framework for the feasibility of HAD.

Nevertheless, it is difficult to improve the generalization of deep-learning-based HAD through the data augmentation methods because of unavailable prior knowledge, complicated imaging environment, and low spatial resolution [34]. Fortunately, Wang *et al.* [35] investigated the generalization behaviors of convolutional neural networks (CNN) from a data perspective and demonstrated the inductive bias of the CNN model on high-frequency components (HFCs) and low-frequency components (LFCs). For example, the LFC is much more "generalizable" than the HFC in the original label case, while the HFC may play a significant role in depicting some objects, especially the smaller ones. Moreover, given some specific information in the HFC, such as object boundaries, can distinguish different objects more effectively, extracting the HFC as a separate processing branch can significantly boost the performance [36]. Accordingly, detecting anomalies in the frequency domain may be more suitable for small and sparse hyperspectral anomalies.

With these in mind, we propose a dual-frequency autoencoder (DFAE) detection model, in which the original HSI is transformed into HFCs and LFCs before detection. Learning in the frequency domain has been applied to multiple image understanding tasks [37], [38], also in the field of hyperspectral pan-sharpening [39]. But it has not been adopted in HAD. Unlike natural images, HSI contains a wealth of spectral information in addition to spatial information. Its continuous smooth part in the spatial domain, which is known as the LFC in natural images, also indicates the similarity of spectral characteristics among neighboring pixels. However, due to varied atmospheric environment and imperfect equipment, the original hyperspectral data have the spectral variation problem, that is, the same material may have dissimilar spectral curves. This phenomenon may negatively impact the performance of HAD, leading to a high false alarm rate or loss of detection accuracy, which is ignored by the existing methods. Different from previous work, we propose spectral rectification to solve this problem and obtain the homogenized LFC while preserving the local similarity of HSI. Meanwhile, we use the Laplacian operator to extract the HFC. In this way, an HSI is transformed into the HFCs and LFCs. For the detection step, using existing deep-learning-based methods has shown that anomalies can be distinguished from the background not only in the low-dimensional latent space but also in the reconstruction space. Thus, we select the typical generative model AE as our basic framework to detect anomalies.

The main contributions of our work are as follows.

1) For the first time, learning in the frequency domain is introduced into HAD, which not only complements specific information to provide a better performance gain but also breaks the dilemma of limited generalization of deep-learning-based methods in sample-free HAD task.
2) To effectively separate the HFCs and LFCs, a spectral rectification method is proposed to generate the LFCs given the local similarity of HSI. Besides, the Laplacian operator with more sensitivity to details is introduced to extract the HFCs.
3) Designed specifically for frequency components of HSI, a novel DFAE architecture processes the two frequency components in parallel, which detects anomalies from low-dimensional representation space and reconstruction space simultaneously.
4) Most deep-learning-based HAD methods require retraining for different datasets because the network model has distinct sensitivity in the image domain. However, our DFAE method detects anomalies in the frequency domain, allowing training on one HSI and testing on others captured from the same sensor. Experiments conducted on real datasets, including both point and structural anomalies, demonstrate its superiority.

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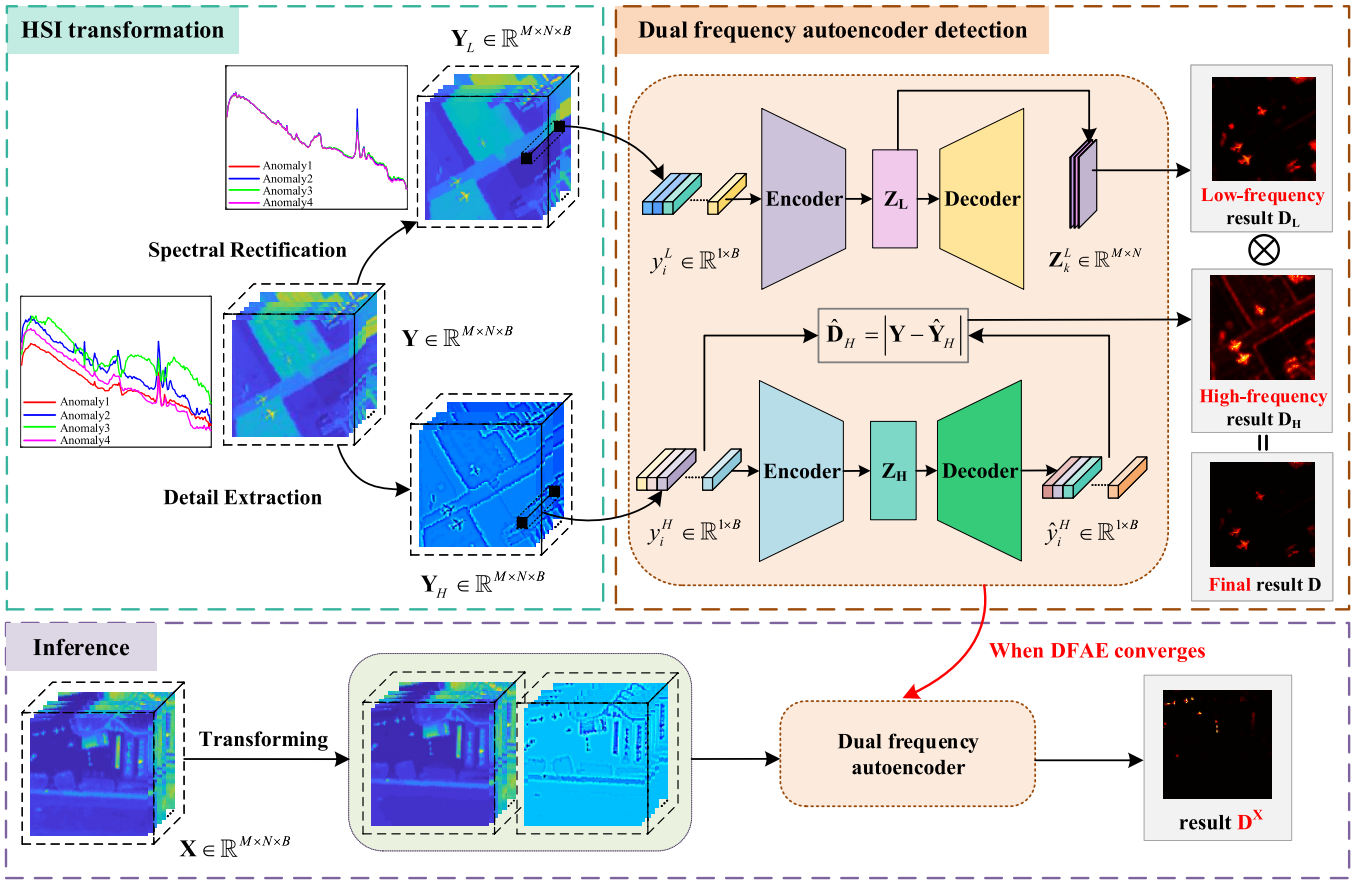


Fig. 1. Framework of the proposed DFAE method in transformed HSI.

The remainder of this article is organized as follows. The proposed DFAE framework is detailed in Section II. Section III presents the experimental results and analysis. Finally, conclusions are drawn in Section IV.

II. Proposed Method

Fig. 1 shows the framework of our proposed DFAE method, which contains three parts: HSI transformation, DFAE detection, and inference on another testing HSI. Let $\mathbf{Y} = [y_1, y_2, \ldots, y_{M\times N}]$ denote an original HSI with $M \times N$ pixels, where $y_i \in \mathbb{R}^{B\times 1}$ stands for a B dimensional spectral vector of the i th pixel. For a given $\mathbf{Y} \in \mathbb{R}^{B\times M\times N}$, the objective of the proposed DFAE method is to design a detection model φ such that

$$\mathbf{D} = \varphi(\mathbf{Y}_L, \mathbf{Y}_H) \tag{1}$$

where $\mathbf{Y}_L$ and $\mathbf{Y}_H$ represent the LFCs and HFCs of $\mathbf{Y}$, respectively. In an ideal case, $\mathbf{Y}_L$ and $\mathbf{Y}_H$ can exactly reconstruct $\mathbf{Y}$, i.e., $\mathbf{Y} = \mathbf{Y}_L + \mathbf{Y}_H$. However, in real-world applications, due to the limitation of imaging sensor and the influence of environment, the original hyperspectral data are always corrupted by noise. Therefore, it is significant to find an appropriate approach to acquire a meaningful $\mathbf{Y}_H$. Besides, how to attain a continuously smoothing $\mathbf{Y}_L$ is also one of the major challenges in our method.

A. HSI Transformation

Under the context of nature images, the term frequency refers to the variation in pixel values over space (i.e., the height and width of the image) [40]. The HFC implies a higher change of values which happens when there is a sharp contrast in the image, such as the object boundary. While the LFC means a smaller change in pixel values which corresponds to the continuous and smooth part of the image. Correspondingly, for HSI, the continuous smooth part in the spatial domain forms its LFC, which also indicates that the spectral characteristics between adjacent pixels are similar. However, the phenomenon that the same object in practical situations cannot guarantee identical spectra may destroy the local similarity of HSI to some extent. Hence, we perform spectral rectification to generate a homogenized $\mathbf{Y}_L$.

1) Spectral Rectification: It is important to make the spectra of the same object as similar as possible while ensuring that the rectified image $\mathbf{Y}_L$ does not violate the physical properties of the original HSI $\mathbf{Y}$. Based on this understanding, we formulate the spectral rectification process as the following minimization problem:

$$\min_{\mathbf{Y}_L} \|\mathbf{Y}_L - \mathbf{Y}\|_F^2 + \alpha \sum_i \sum_{d\in\{h,v\}} \frac{\left|\nabla_d y_i^L\right|}{\max\left(\left|\nabla_d y_i^L\right|, \varepsilon\right)} \tag{2}$$

where α is a nonnegative coefficient set to 0.001 to balance the two involved terms, and $|\nabla_d y_i^L|$ represents the spectral

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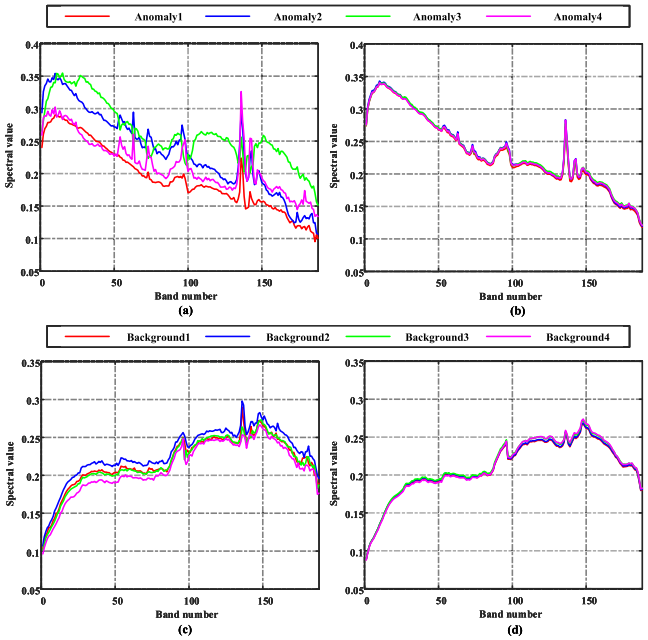


Fig. 2. Comparison of the rectified and original images in the San Diego dataset. (a) Spectral curves of anomalies in the original HSI. (b) Spectral curves of anomalies in the rectified HSI. (c) Spectral curves of background samples in the original HSI. (d) Spectral curves of background samples in the rectified HSI.

gap of the i th pixel in the rectified image $\mathbf{Y}_L$ with forward difference in the d direction (i.e., horizontal direction h or vertical direction v). In addition, the small positive constant ε is introduced to determine the spectral gap's degree and set to 0.01. When the spectral gap $|\nabla_d y_i^L|$ is greater than ε, i.e., $|\nabla_d y_i^L| > \varepsilon$, the weight coefficient $(1/|\nabla_d y_i^L|)$ is small enough to reduce the difference, whereas the weight coefficient turns to be large when $|\nabla_d y_i^L|$ is smaller than ε. In summary, the former $\|\mathbf{Y}_L - \mathbf{Y}\|_F^2$ in (2) is the fidelity term to prevent $\mathbf{Y}_L$ from deviating greatly from $\mathbf{Y}$, while the latter is to achieve rectification by narrowing the larger spectral gaps and retaining the smaller gaps between adjacent pixels.

According to [41], (2) can be rewritten as the following matrix form:

$$\min_{\mathbf{Y}_L} \ (\mathbf{Y}_L - \mathbf{Y})^T(\mathbf{Y}_L - \mathbf{Y}) + \alpha \mathbf{Y}_L^T \mathbf{L}_g \mathbf{Y}_L$$
$$\mathbf{L}_g = \sum_{d\in\{h,v\}} \mathbf{D}_d^T \mathbf{A}_d \mathbf{A}_d \mathbf{D}_d. \quad (3)$$

Here, $\mathbf{L}_g$ is a symmetric positive sparse matrix, where $\mathbf{A}_d$ is the diagonal matrix with the i th diagonal entries being $(1/\max(|\nabla_d y_i^L|, \varepsilon))$, and the matrix $\mathbf{D}_d$ is a discrete differentiation operator. Since (3) is quadratic, it has a unique minimum that can be solved from a linear system with the sparse matrix

$$\left(\mathbf{I} + \alpha \mathbf{L}_g\right)\mathbf{Y}_L = \mathbf{Y}. \quad (4)$$

In our implementation, the preconditioned conjugate gradient (PCG)-based iterative solver is used to calculate the solution $\mathbf{Y}_L$ after construction of a five-point spatially inhomogeneous Laplacian matrix $\mathbf{L}_g$ which is generated by the method in [42].

Consequently, the spectral curves of the background and anomalies become more uniform, as shown in Fig. 2.

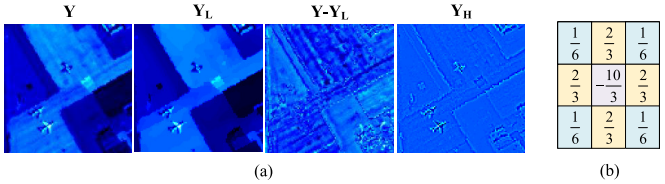


Fig. 3. (a) Visual comparison of HSI transformation. (b) Filter mask of the Laplacian operator.

This means that the spectral rectification process successfully alleviates the spectral variation problem, which can effectively preserve the local similarity of HSIs and achieve the separation of LFCs $\mathbf{Y}_L$.

2) Detail Extraction: We adopt the Laplacian operator to extract the HFCs $\mathbf{Y}_H$ instead of directly subtracting $\mathbf{Y}_L$ from the original image (i.e., $\mathbf{Y} - \mathbf{Y}_L$), since the result of direct subtraction will keep all the noise and have almost no anomaly information, as illustrated in Fig. 3(a).

Formally, this process can be expressed as

$$\mathbf{Y}_j^H = \mathrm{Lap}(\mathbf{Y}_j), \quad j = 1, 2, \ldots, B \quad (5)$$

where $\mathbf{Y}_j^H \in \mathbb{R}^{M\times N}$ represents a 2-D image of size $M \times N$ on the j th band of the HFCs $\mathbf{Y}_H = \left[\mathbf{Y}_1^H; \mathbf{Y}_2^H; \ldots; \mathbf{Y}_B^H\right]$. The Laplacian operator is a second derivative operator, which is more sensitive to details than the first derivative operator [43]. To better explain the process of obtaining $\mathbf{Y}_H$, the filter mask of the Laplacian operator in our method is illustrated in Fig. 3(b). With this filter mask, the Laplacian operator can process high-frequency information in any direction and respond more strongly to isolated pixels, which always appear in anomalous targets or boundaries.

As discussed in our model (1), once the original HSI is transformed into $\mathbf{Y}_L$ and $\mathbf{Y}_H$, the detection model φ can start learning in the form of unsupervised learning through DFAE.

B. Dual-Frequency Autoencoder Detection

1) Network Architecture: Considering the model (1) is desired to be able to process two different information at the same time, we construct the dual AE architecture to process the above-mentioned $\mathbf{Y}_L$ and $\mathbf{Y}_H$ in parallel.

In the low-frequency AE, the spectral vector $y_i^L \in \mathbb{R}^{B\times 1}$ is regarded as a training sample, and the collection of all spectral vectors is the training examples' set, i.e., $\mathbf{Y}_L = \left[y_1^L, y_2^L, \ldots, y_{M\times N}^L\right]$, which is fed into the network. The reconstructed result $\hat{\mathbf{Y}}_L$ is the output, and $\mathbf{Z}_L$ represents the latent feature of $\mathbf{Y}_L$. Thus, the low-frequency encoding process and decoding process are demonstrated as

$$\mathbf{Z}_L = \sigma\left(\mathbf{W}_L^e \mathbf{Y}_L + b_L^e\right) \quad (6)$$
$$\hat{\mathbf{Y}}_L = \sigma\left(\mathbf{W}_L^d \mathbf{Z}_L + b_L^d\right) \quad (7)$$

where the parameters $\mathbf{W}_L^e$ and b_L^e denote the weight matrix and the bias vector of the encoding layers, respectively, whereas the parameters $\mathbf{W}_L^d$ and b_L^d represent the weight matrix and the bias vector of the decoding layers, respectively.

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Similarly, in the high-frequency AE, $\mathbf{Y}_H$ and $\hat{\mathbf{Y}}_H$ stand for the input and the output, respectively. The hidden representation of $\mathbf{Y}_H$ is denoted by $\mathbf{Z}_H$. Hence, we illustrate the high-frequency encoding and decoding process as

$$\mathbf{Z}_H = \sigma\left(\mathbf{W}_H^e \mathbf{Y}_H + b_H^e\right) \tag{8}$$

$$\hat{\mathbf{Y}}_H = \sigma\left(\mathbf{W}_H^d \mathbf{Z}_H + b_H^d\right) \tag{9}$$

where the parameters $\mathbf{W}_H^e, b_H^e$ and $\mathbf{W}_H^d, b_H^d$ represent the weight matrix and the bias vector of the encoding and decoding layers, respectively. Furthermore, the activation function σ of each architecture is the sigmoid function

$$\sigma(q) = \frac{1}{1+\exp(-q)} \tag{10}$$

so that the output value range of the function is limited to 0–1, i.e., $0 \le \sigma(q) \le 1$.

In general, the parameters $\theta = \left\{\mathbf{W}_L^e, b_L^e, \mathbf{W}_L^d, b_L^d\right\}$ and $\hat{\theta} = \left\{\mathbf{W}_H^e, b_H^e, \mathbf{W}_H^d, b_H^d\right\}$ are optimized by minimizing the mean square error (MSE) between the reconstruction result and the original input. However, this kind of AE network with MSE may have over-fitting problems due to its over-strong representation ability. To tackle this problem, we add a Kullback-Leibler (KL)-divergence-based sparsity constraint as a penalty term in the loss function, which is motivated by sparse AE [44], to facilitate AE discovering more latent structure in the data.

Specifically, the sparsity constraint is imposed on the hidden units. Let $\hat{\rho}_k$ denote the actual average activation of the hidden unit k, which is defined as

$$\hat{\rho}_k = \frac{1}{MN}\sum_{i=1}^{MN} \mathbf{Z}_k^F\left(y_i^F\right), \quad F \in \{L, H\} \tag{11}$$

where $\mathbf{Z}_k^F\left(y_i^F\right)$ denotes the activation of the unit $\mathbf{Z}_k^F$ when the sample $y_i^F \in \mathbb{R}^{B\times 1}$ is given, and the superscript F represents this equation is applied to both the AEs. The smaller the value of $\hat{\rho}_k$, the sparser the AE network.

Therefore, the complete loss function of each AE is

$$L_F = \frac{1}{MN}\left\|\mathbf{Y}_F - \hat{\mathbf{Y}}_F\right\|^2 + \beta\sum_{k=1}^{K}\mathrm{KL}(\rho \parallel \hat{\rho}_k), \quad F \in \{L, H\}. \tag{12}$$

Here, β controls the influence of KL divergence, while ρ denotes the expected average activation of the hidden unit and is set to 0.02, and K is the number of units in the hidden layer, which is restricted to 3 and 60 for the low-frequency and high-frequency AEs, respectively. More details about these parameters will be displayed in Section III. Consequently, the training process of DFAE is to optimize the learnable parameters θ and $\hat{\theta}$ by minimizing the loss function (12).

2) Background Removal: With the parameters θ and $\hat{\theta}$ being well-optimized, both the inputs $\mathbf{Y}_L$ and $\mathbf{Y}_H$ can be well-represented by latent features $\mathbf{Z}_L$ and $\mathbf{Z}_H$, and the corresponding reconstructed HSIs $\hat{\mathbf{Y}}_L$ and $\hat{\mathbf{Y}}_H$ are generated. Inspired by our previous works [20], [32], anomalies can be detected by removing the background after enhancing the discrimination between the background and anomalies.

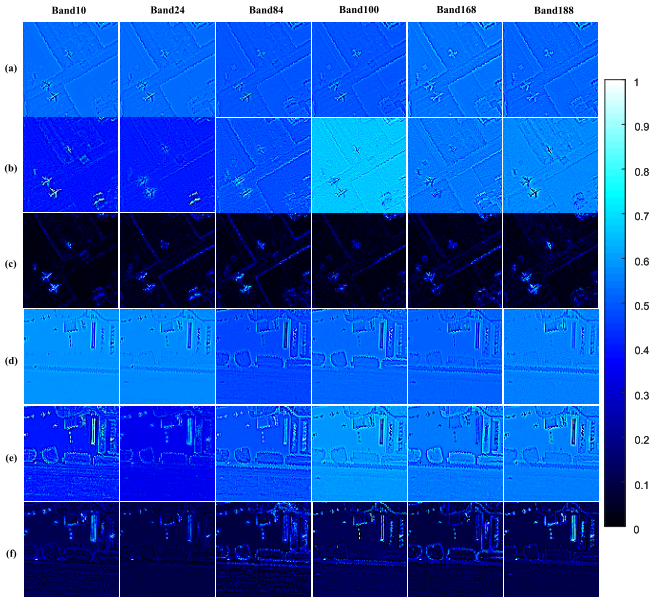


Fig. 4. Learning results of the high-frequency AE on different bands in visual. (a) HFCs $\mathbf{Y}_H$. (b) Reconstructed result $\hat{\mathbf{Y}}_H$. (c) Residual image $\hat{\mathbf{D}}_H$. (d) HFCs $\mathbf{X}_H$. (e) Result $\hat{\mathbf{X}}_H$. (f) Residual image $\hat{\mathbf{D}}_H^{\mathbf{X}}$.

According to the different characteristics of the input data, we choose different background removal methods.

a) The LFCs: $\mathbf{Y}_L$ has strong relevance among pixels of the same object, and anomalous targets usually appear in much smaller areas than the background. Based on these facts, we use the morphological attribute filter [45] and difference operation to the latent features $\mathbf{Z}_L$ to remove the background. The morphological opening operation $A^O\left(\mathbf{Z}_k^L\right)$ and the closing operation $A^C\left(\mathbf{Z}_k^L\right)$ are applied to remove the bright and dark connected components when the area $\le \eta$. Then the differential map between the filtered image and the original latent features is calculated as (13) to preserve all small area components, most of which are anomalies

$$\hat{\mathbf{D}}_L\left(\mathbf{Z}_k^L\right) = \left|\mathbf{Z}_k^L - A^O\left(\mathbf{Z}_k^L\right)\right| + \left|\mathbf{Z}_k^L - A^C\left(\mathbf{Z}_k^L\right)\right|. \tag{13}$$

b) The HFCs: $\mathbf{Y}_H$ is more highlighted at anomalies or boundaries compared with the background. However, the background samples occupy the majority of all training samples, allowing them to recover better in the reconstructed result $\hat{\mathbf{Y}}_H$. The anomalies, in contrast, fail to recover the original features in $\hat{\mathbf{Y}}_H$ since the network cannot learn the characteristics of abnormal targets sufficiently with a small number of samples. Hence, the reconstruction residual image $\hat{\mathbf{D}}_H = \left|\mathbf{Y}_H - \hat{\mathbf{Y}}_H\right|$ refers to the image in which the anomalies are separated from the background, as shown in Fig. 4(c) and (f).

For the convenience of subsequent results' combination, a simple fusion strategy is conducted to unify dimension, and the guided filter [46] is applied to further refine the fused image

$$\mathbf{D}_F = \mathrm{GIF}\left(\frac{1}{S}\sum_{s=1}^{S}\hat{\mathbf{D}}_F\right), \quad F \in \{L, H\}. \tag{14}$$

Here, S stands for the number of bands in $\hat{\mathbf{D}}_L$ and $\hat{\mathbf{D}}_H$, where $\hat{\mathbf{D}}_L$ is 3 and $\hat{\mathbf{D}}_H$ is B.

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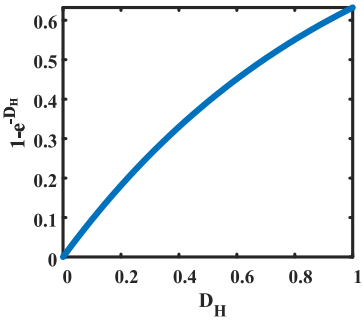


Fig. 5. Shape of the exponential constrained nonlinear function.

3) Results Combination: The low-frequency result $\mathbf{D}_L$ and the high-frequency result $\mathbf{D}_H$ can provide complementary information. There are two situations as follows. (1) In $\mathbf{D}_H$, the difference between the pixel values of anomalies and false alarms is large, but in $\mathbf{D}_L$ the two values are close. (2) The pixel values of anomalies and false alarms are similar in $\mathbf{D}_H$, while they are quite different in $\mathbf{D}_L$. To better combine the advantages of these two detection results, we perform the exponential constrained nonlinear function f

$$\mathbf{D} = f(\mathbf{D}_H) \cdot \mathbf{D}_L = \left(1 - \mathrm{e}^{-\mathbf{D}_H}\right) \cdot \mathbf{D}_L. \tag{15}$$

According to the schematic of the nonlinear function plotted in Fig. 5, the pixel value in $\mathbf{D}_H$ will be suppressed to varying degrees. In situation (1), $f(\mathbf{D}_H)$ can help eliminate the false alarms in $\mathbf{D}_L$, because the pixel value at the same position is almost 0 in $f(\mathbf{D}_H)$. In situation (2), our combination strategy can prevent the false alarms in $\mathbf{D}_L$ from degrading accuracy due to multiplying by a larger value. Therefore, using the proposed combination strategy can get better detection results than direct multiplication. More details about the high- and low-frequency results will be analyzed in Section III-C.

C. Inference

Unlike the existing HAD methods based on unsupervised learning, our proposed DFAE method can detect different HSI datasets captured from the same sensor using the well-optimized parameters learned from an HSI. In other words, for another HSI $\mathbf{X}$, anomalies can be detected by the trained model φ of the parameters (i.e., θ and $\hat{\theta}$) optimized by the original HSI $\mathbf{Y}$. That is

$$\mathbf{D}^{\mathbf{X}} = \varphi\left(\mathbf{X}\middle|\theta, \hat{\theta}\right). \tag{16}$$

Fig. 4 illustrates some results of the high-frequency AE to promote the understanding of the generalization of our detection model φ. For anomaly, there are significant differences between the reconstructed image and the input image. Comparing Fig. 4(a) and (b), some airplanes with darker colors in the input image $\mathbf{Y}_H$ become prominent in the reconstructed image $\hat{\mathbf{Y}}_H$, such as bands 10 and 84, while some airplanes with brighter colors in $\mathbf{Y}_H$ are darkened in $\hat{\mathbf{Y}}_H$ but have highlighted edges, like band 188. The background changes in all these bands are relatively small. Correspondingly, in the test dataset $\mathbf{X}$, whose visual results are displayed in Fig. 4(d) and (e), the changes in the reconstructed image $\hat{\mathbf{X}}_H$ relative to the input image $\mathbf{X}_H$ are approximately the same as the changes in the learned dataset, such as bands 10, 84, and 188. Therefore, the conclusion can be drawn that the determined model φ learned by a certain HSI can be generalized for anomaly detection from other HSI datasets.

Moreover, the limitation of inference is that the number of bands in the test datasets must be the same as the number of bands in the learned dataset, since the input of dual AE is spectral vectors. This can be achieved by the removal of bands with low signal-to-noise ratios.

III. Experiments

In this section, the proposed DFAE method is validated on four real HSIs captured over different scenes. In these images, some anomalies appear as points while others are in the form of areas with structural information, which are detailed as follows.

San Diego Dataset: The first dataset is a widely used HSI collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the San Diego airport area, CA, USA. The whole original image scene contains 400 × 400 pixels with 224 spectral channels. After removing the water vapor absorption and noisy bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands with wavelengths ranging from 0.4 to 2.5 μm are retained. A region with a size of 100 × 100 pixels is selected from the center of the whole image. Three airplanes with structural information are anomalous targets of interest, consisting of 134 pixels with a spatial resolution of 3.5 m.

Texas Coast Dataset: The second dataset is a scene of the Texas coast area captured by the AVIRIS sensor on August 29, 2010. It consists of 100 × 100 pixels, and the spatial resolution is 17.2 m per pixel. This dataset has 204 spectral bands varying from 0.45 to 1.35 μm, some of which are corrupted by a series of strip noise, so it is considered a challenging dataset. The houses occupying 67 pixels are anomalies to be detected, which also have structural information.

Gainesville Dataset: The third dataset was acquired by the AVIRIS sensor from Gainesville city, North Florida, USA. The image scene covers an area of 100 × 100 pixels, with 191 spectral bands and a 3.5-m spatial resolution. There are ten anomalous targets of 52 pixels in total, accounting for 0.52% of the image, which can be regarded as point anomaly separately.

Cat Island Dataset: The fourth dataset was recorded by the AVIRIS sensor over the area of Cat Island. It has 188 spectral bands and 150 × 150 pixels with a spatial resolution of 17.2 m. Sea and an island make up the background, while a ship on the sea is regarded as an anomaly. This target is composed of 19 pixels and accounts for 0.08% of the whole image, which appears in the form of points.

To evaluate and verify the **generalization** of our DFAE method, we use the most common San Diego Dataset as the training set to learn the optimal parameters, and the remaining datasets are used as the testing set for anomaly detection by the determined model φ with the well-learned parameters,

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TABLE I
DETAILED INFORMATION FOR EXPERIMENTAL DATASETS

Function	Datasets	Sizes	Selected bands	Anomaly size	Anomaly type	Resolution
Training	San Diego	100×100×189	1-188	134 pixels	structural anomaly	3.5 m
Testing	Texas Coast	100×100×204	1-188	67 pixels	structural anomaly	17.2 m
Testing	Gainesville	100×100×191	1-188	52 pixels	point anomaly	3.5 m
Testing	Cat Island	150×150×188	1-188	19 pixels	point anomaly	17.2 m

which means that all the detection results in this article are determined by a fixed set of parameters. It is important to note that all the datasets for experiments need to have the same number of bands. To use more datasets for performance evaluation, we choose the least number of bands 188 in all the datasets as standard. The bands selected of the experimental datasets are listed in Table I.

A. Experimental Setup

1) Evaluation Metrics: In the experiments, the receiver operating characteristic (ROC) [47] curve and the area under the curve (AUC) [48] are adopted to evaluate the detection performance of the proposed method and its comparison methods. The ROC curve is plotted by true positive rate (TPR) versus false positive rate (FPR) at various thresholds (τ), which can qualitatively determine the superiority among different methods under the same false alarm condition. The AUC value obtained by calculating the whole area under the ROC curve can quantitatively illustrate the detection performance. The closer the AUC value of (TPR, FPR) is to 1, the better the detection accuracy. Correspondingly, the smaller the AUC value of (FPR, τ), the lower the probability of false detection. Furthermore, we also use the Box–Whisker plots [49] to evaluate the separability between anomaly and background. The boxes in the Box–Whisker plot describe the distribution range of detection values of the anomaly and background. Thus, the relative distance of these two boxes can measure the anomaly–background separability.

2) Comparison Methods: Six HAD methods are used for comparison, including LSDM-mixture of Gaussian (MOG) [17], PTA [21], FrFE [22], subspace selection-based discriminative forest (SSDF) [23], orthogonal subspace-projection (OSP)-GoDec [50], and AAE [51]. LSDM-MOG is an anomaly detection method based on RPCA, of which the parameter settings are discussed in [17]. The latest PTA method decomposes HSI into a background tensor and an anomaly tensor. According to [21], the best truncated rank is smaller than 20, for which we select the best value for each dataset by traversing from 0 to 20. The FrFE-based method also detects anomalies in a transform domain, which is similar to the proposed method. Its optimal parameters are shown in [22]. The SSDF performs HAD with the help of the concept of isolation, and its parameters "the number of trees", "the number of selected bands," and "the subsample number" are set to 500, 40, and 2500 or 2000, respectively, according to [23]. In [50], there are 12 types of anomaly detectors derived from OSP-GoDec. We select the best detection result for each experimental dataset for comparison, and the parameter "false alarm ratio" is set to 0.01. In addition, the AAE algorithm is also compared to show the advantages of DFAE over the general deep-learning-based method.

3) Implementation Details: As stated in Section II-B1, our network architecture consists of two AEs. In the low-frequency AE, the encoder and the decoder adopt the four fully connected layers, which can be $E\{B, B/2, B/4, 3\}$ and $D\{3, B/4, B/2, B\}$. Here, B is the number of bands of input HSIs, i.e., 188 in our experiments. The size of the latent feature space is restricted to 3 units, which is experiential knowledge presented in [19]. Considering the fact that data distribution of the HFCs is more distinct than the original image, the high-frequency AE has only two fully connected layers to avoid the overfitting problem. Its number of latent features is 60, which will be analyzed in Section III-C. Furthermore, our network is optimized using a stochastic Adam optimizer with 1300 epochs. The learning rates are 0.0001 and 0.0002 for the low-frequency and high-frequency AEs, respectively. All the experiments are carried out on an Intel Core i5-7200 central processing unit (CPU) machine with 16 GB of random access memory (RAM). The learning and inferring processes based on the network model are implemented in Python 3.6.2 and TensorFlow-CPU 1.2.1 environment, while the others are performed in MATLAB R2018b environment.

B. Parameters Analysis

With the configured hyperparameters described above, several parameters mainly impact the performance of DFAE, including the number of hidden nodes in the high-frequency AE (K_H), the coefficient α and constant ϵ in (2), the constant ρ and weight of sparsity constraint (i.e., β_L and β_H) in (12), and the area size η described in Section II-B2. In experiments, when one of the parameters is being analyzed, other parameters are fixed as the set optimal values, which are $K_H = 60$, $\rho = 0.02$, $\beta_L = 0.01$, $\beta_H = 0.0001$, $\alpha = 0.001$, $\epsilon = 0.01$, and $\eta = 55$ or $\eta = 15$. The optimal values of these parameters are estimated by an empirical method, which is evaluated by the AUC value of (TPR, FPR). As illustrated in Fig. 6(a), when the parameter K_H is 60, the proposed method achieves the best performance on most datasets except for the Texas Coast dataset. Nevertheless, the AUC value of the Texas Coast dataset ranks second at $K_H = 60$ and is very close to the maximum. Therefore, it is reasonable to set $K_H = 60$. After determining the network structure, the parameters in loss functions are analyzed. Fig. 6(b) shows the effect of the parameter ρ on the detection performance over four experimental datasets. When the parameter ρ is set to 0.02, the AUC values of all the three testing datasets are maximums, while the AUC value of the training set is the second. Even though the San Diego dataset has the best performance at $\rho = 0.03$, it can be seen from Fig. 6(b) that the

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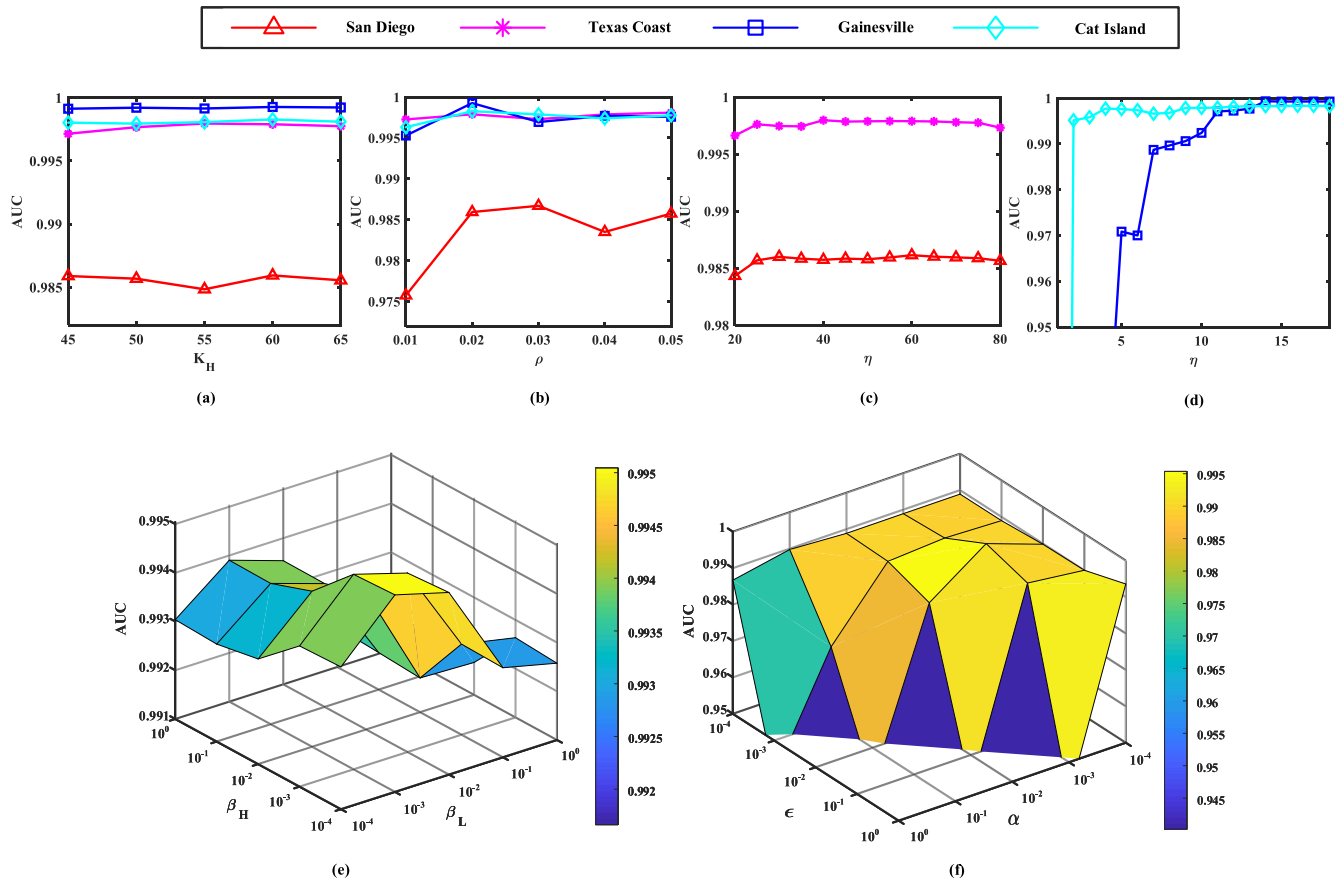


Fig. 6. Analysis of the parameters over experimental datasets. (a) Number of hidden nodes K_H. (b) Constant ρ in loss functions. (c) Area size η on structural anomaly. (d) Area size η on point anomaly. (e) Coefficients β_L and β_H of loss functions. (f) Coefficient α and constant ϵ.

difference in the AUC value of the Cat Island dataset is greater than that of the San Diego dataset between 0.02 and 0.03. Thus, we choose 0.02 as the optimal value of the parameter ρ. As for the coefficients β_L and β_H, we use the average accuracy of all the datasets as the evaluation criterion to ensure the generalization of the detection model. In the 3-D diagram of β_L and β_H, the yellow part indicates that the AUC value is higher, whereas the blue part is lower. Given the color distribution in Fig. 6(e), $\beta_L = 0.01$ and $\beta_H = 0.0001$ are selected as the optimal coefficient values. A similar 3-D diagram reflecting the effect of parameters α and ϵ on average accuracy is plotted in Fig. 6(f). The change of color shows that the value of AUC decreases as the value of parameters α and ϵ increases. For easier visual comparison, some results far less than 0.95 are not shown in the figure. In Fig. 6(f), there is a distinct bright yellow bump at $\alpha = 0.001$ and $\epsilon = 0.01$, which indicates that the optimal values of parameters are $\alpha = 0.001$ and $\epsilon = 0.01$. In addition, due to the different scales of abnormal samples, the area size η should depend on the anomaly type. For structural anomaly, as shown in Fig. 6(c), both the San Diego dataset and Texas Coast dataset get the second-ranked AUC value at 55, although their best AUC values are at 60 and 40, respectively. For point anomaly, as illustrated in Fig. 6(d), the optimal value of the Gainesville dataset is taken at 14, the optimal value of the Cat Island dataset is taken at 16, and both of them are less sensitive when η varies from 14 to 19. Hence, we choose the middle-value $\eta = 15$ for most of the point anomaly datasets.

C. Component Analysis

The proposed detection model $\mathbf{D} = \varphi(\mathbf{Y}_L, \mathbf{Y}_H)$ comprises two parts, the detection of the HFC and the detection of the LFCs, complementing each other. To explore the influence of these two parts on detection performance, we also conduct a comparison experiment for each of them. Therefore, we have three groups containing five detection results for comparative analysis. In the first group, the original HSI is directly fed into the AE for anomaly detection using the reconstruction errors as anomaly scores, and its result, namely, $\mathbf{Y}$_H, is compared with the HFC. The second group consists of the LFC and the detection result of the original HSI without spectral rectification, namely, $\mathbf{Y}$_L. The third group contains the HFC, the LFC, and the final result in our model. The detection maps and quantified AUC values of the above experiments are illustrated in Fig. 7 and Table II, respectively.

Comparing the results of the first group, the following observations are made. First, almost all the background in $\mathbf{Y}$_H is preserved, which makes its AUC value of (FPR, τ) extremely high. Second, $\mathbf{Y}$_H has a poorer discrimination ability for anomalies. As shown in Fig. 7, the shape of the airplane on the upper right in the San Diego dataset is blurry, and the background in the Gainesville and Cat Island datasets

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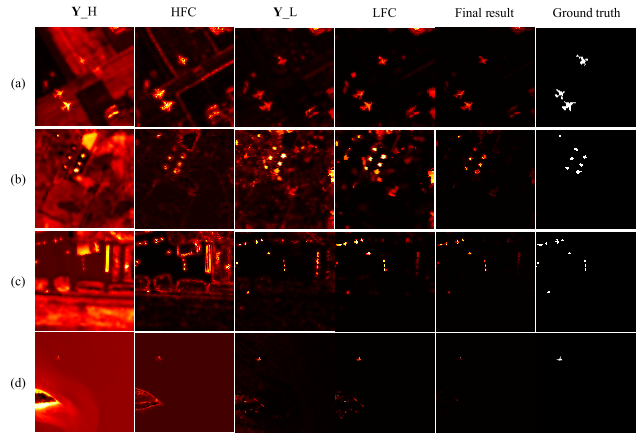


Fig. 7. Detection maps based on different components. (a) San Diego. (b) Texas Coast. (c) Gainesville. (d) Cat Island.

TABLE II
EFFECTIVENESS ANALYSIS OF LFC AND HFC ON EXPERIMENTAL DATASETS

Datasets	AUC Values of (TPR, FPR)				
	Y_H	HFC	**Y**_L	LFC	Final
San Diego	0.97841	0.98437	0.98473	0.98149	**0.98594**
Texas Coast	0.93814	0.98616	0.99573	0.99730	**0.99790**
Gainesville	0.73176	0.95607	0.99311	0.99922	**0.99927**
Cat Island	0.84453	0.92358	0.98818	0.99765	**0.99828**
Average	0.87321	0.96255	0.99044	0.99392	**0.99535**
Datasets	AUC Values of (FPR, τ)				
	Y_H	HFC	**Y**_L	LFC	Final
San Diego	0.14177	0.06413	0.03037	0.01655	**0.00637**
Texas Coast	0.18880	0.04779	0.08543	0.03445	**0.00611**
Gainesville	0.25295	0.07747	0.02227	0.00677	**0.00272**
Cat Island	0.24138	0.09816	0.01533	0.00236	**0.00110**
Average	0.20623	0.07189	0.03835	0.01503	**0.00408**

is more outstanding than the anomaly. Most importantly, the generalization ability of **Y**_H is limited, especially when the anomaly type of the training set is different from that of the testing sets. As for the second group, due to the lack of a spectral rectification process, **Y**_L is generally more disturbed by the background and noise, which causes its AUC value of (FPR, τ) more than twice higher than that of LFC and affects the detection accuracy to a certain extent.

The third group is used to specifically analyze the impact of HFC and LFC on the final result. For the final result of the San Diego dataset, HFC provides more prominent abnormal targets, while LFC provides a lower false alarm rate. The opposite is true in the Texas Coast dataset, which has a higher AUC value of (TPR, FPR) for the LFC and purer background for the HFC. Although the LFC of the Gainesville dataset has a fairly high AUC value of (TPR, FPR), there are several false detections in the upper right corner of its detection map that are obscured in the final result for combining with the HFC. The final result of the Cat Island dataset minimizes the false alarm rate of the island without loss of detection accuracy.

To further validate the superiority of detecting anomalies in the frequency domain, we replaced our detection model with the RX detector. In other words, LFC and HFC are fed into the RX detector separately, and their detection results are combined by (15). Taking the AUC values in Table III, the

TABLE III
EFFECTIVENESS ANALYSIS OF LFC AND HFC ON EXPERIMENTAL DATASETS USING THE RX DETECTOR

Datasets	AUC Values of (TPR, FPR)			
	RX	LFC_RX	HFC_RX	Final_RX
San Diego	0.94039	0.96239	0.93861	**0.97027**
Texas Coast	0.99120	0.99356	0.98940	**0.99539**
Gainesville	0.95132	**0.97867**	0.89618	0.97680
Cat Island	0.98072	0.98471	0.99319	**0.99402**
Average	0.96591	0.97983	0.95435	**0.98412**
Datasets	AUC Values of (FPR, τ)			
	RX	LFC_RX	HFC_RX	Final_RX
San Diego	0.05974	0.03064	0.09040	**0.00734**
Texas Coast	0.05393	0.02372	0.08359	**0.00518**
Gainesville	0.03555	0.02774	0.07481	**0.00638**
Cat Island	0.00751	0.00806	0.01087	**0.00224**
Average	0.03918	0.02254	0.06492	**0.00529**

performance of the RX detector on LFC is much better than on the original image both in detection accuracy and false alarm, which verifies that the spectral rectification process we proposed is essential. Moreover, except for the Gainesville dataset, the detection performance of LFC is further improved with the help of HFC. In particular, the HFC of the Cat Island dataset obtains the highest AUC value of (TPR, FPR) compared with the LFC and the original HSI, which implies that the anomaly is mainly contained in HFC, and HFC can better find anomaly objects with simple scenarios. In summary, both HFC and LFC play a significant role in HAD, and the combination achieves a satisfactory result.

D. Detection Effectiveness and Efficiency Assessment

The visual detection maps for all the experimental datasets are illustrated in Fig. 8, and the corresponding ROC curves are shown in Fig. 9. Table IV calculates the AUC values for the compared methods, in which the best and second results are highlighted and underlined.

First, by observing the visual comparisons in Fig. 8, we note that the proposed DFAE method can separate the anomalies from the background. Other methods are more likely to fail to distinguish anomaly objects or have high false detection with interference from background and noise. In other words, the DFAE method is always the closest to the ground truth than other comparison methods. Another notable advantage is that for different datasets, our DFAE method only needs to tune the area size parameter to the corresponding value of the point anomaly or structure anomaly, and stable and acceptable performances can be obtained on most datasets.

Concretely, in Fig. 8(a), the DFAE method can well-detect three airplanes and suppress the background with rich categories. The OSP-GoDec algorithm has a similar detection map to the DFAE method, but the detection results of two airplanes are less evident. Although the FrFE algorithm finds obvious aircraft targets and obtains the second-ranked AUC value of (TPR, FPR), it mistakes the hangar's roof as the target of interest. The MOG method not only misidentifies the roof as the anomaly to be detected but also the airplane found on the upper right is indistinguishable because it is confused in the background. A similar problem also arises in the SSDF method. Many backgrounds are being false detections in the SSDF, and the airplane in the lower left is confused in false

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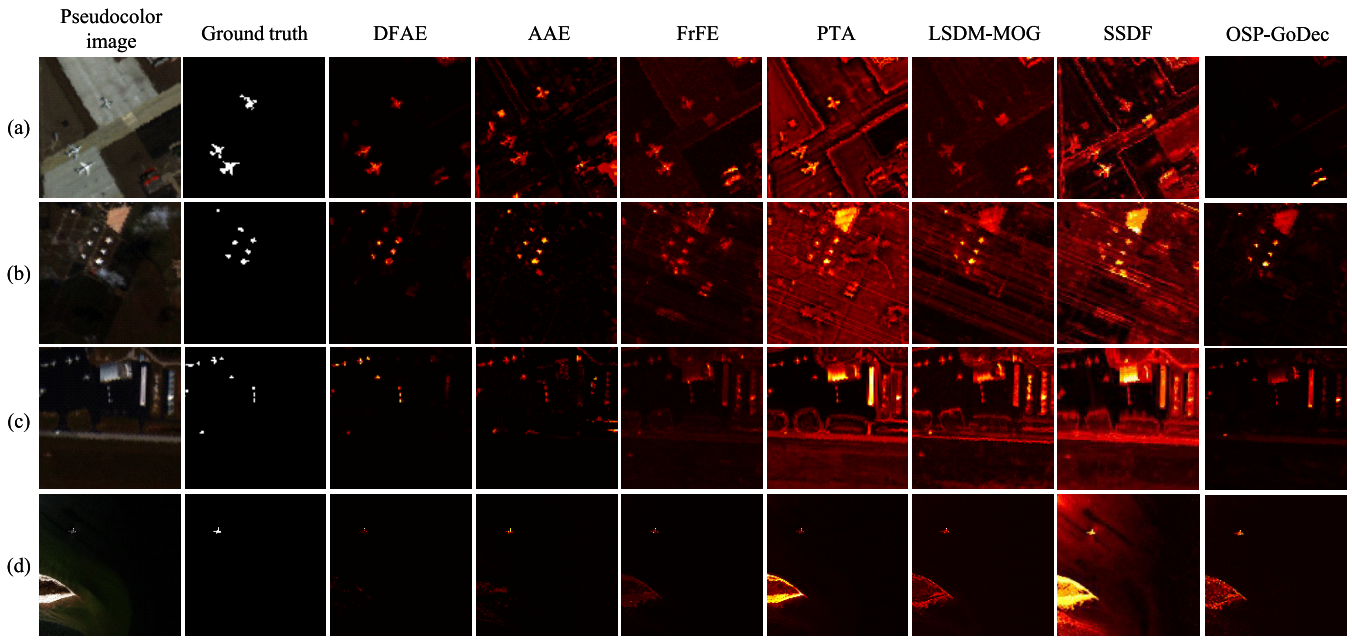


Fig. 8. Pseudo-color image, ground truth, and detection maps of the compared methods for (a) San Diego, (b) Texas Coast, (c) Gainesville, and (d) Cat Island.

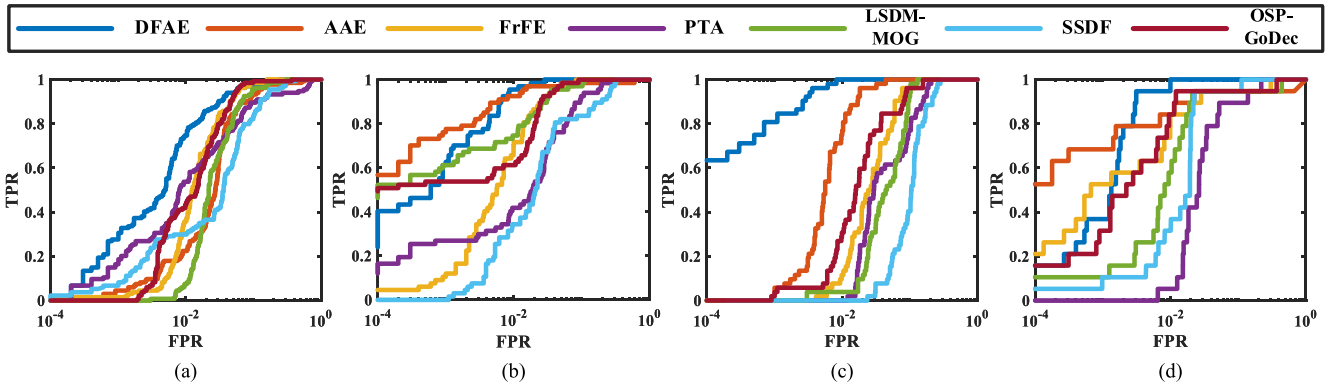


Fig. 9. ROC curves of the compared methods on (a) San Diego, (b) Texas Coast, (c) Gainesville, and (d) Cat Island.

detections. The AAE and PTA methods can detect complete aircraft shape, but they are susceptible to background and noise, degrading the detection accuracy as shown in Table IV. As for the Texas Coast dataset in Fig. 8(b), our DFAE method not only eliminates the strip noise but also removes the background that is prone to misdetection. For instance, the clouds are falsely detected in the AAE method. Like the DFAE and AAE methods, the detection map of the OSP-GoDec algorithm is not affected by the strip noise. However, it falsely detected a block of a meadow as the target of interest. The other four methods, FrFE, PTA, MOG, and SSDF, are all affected by strip noise to varying degrees. Among them, FrFE is the least affected, showing the advantages of detection in the transform domain. In the scenario of point anomalies with complex backgrounds, as shown in Fig. 8(c), our method still does a great job of detecting anomaly objects, which are almost free of background interference. In contrast, the AAE, MOG, and OSP-GoDec methods always have several false detections on the right or bottom of the detection maps while finding abnormal samples. In the FrFE algorithms, due to insufficient contrast between the anomaly and the background, it is hard to recognize the point anomalies. In the PTA and SSDF methods, the background is even more prominent than the anomaly objects, resulting in a low AUC value of (TPR, FPR). Taking Fig. 8(d), the island is an inherent false detection among the six compared methods. Only our DFAE method can correctly distinguish the island and the ship considered as the anomaly. Meanwhile, the AUC values of (TPR, FPR) and (FPR, τ) listed in Table IV quantitatively illustrate that our proposed DFAE method exhibits competitive performance compared with other advanced methods, as it achieves optimal both in high detection accuracy and low false alarm.

Furthermore, Fig. 10 displays the Box–Whisker Plots corresponding to the detection results of different comparison methods. The red and blue boxes represent the distribution of anomaly and background for each HAD method, respectively.

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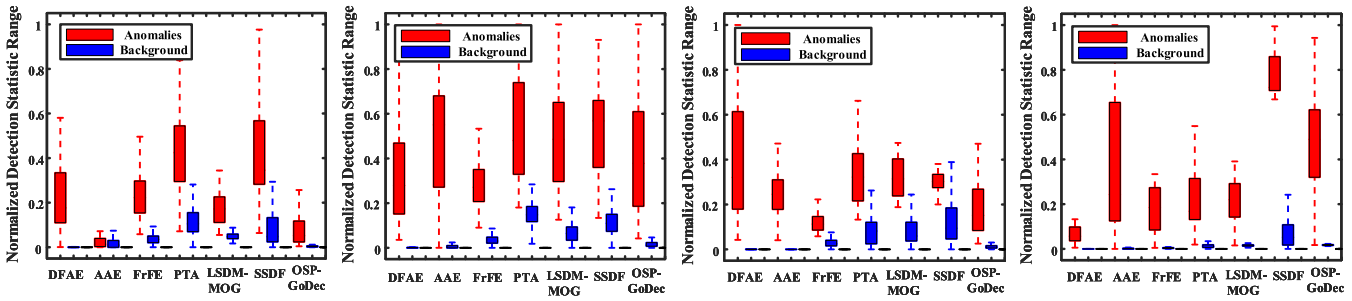


Fig. 10. Box–Whisker Plots of the compared methods on (a) San Diego, (b) Texas Coast, (c) Gainesville, and (d) Cat Island.

TABLE IV
AUC VALUES OF THE COMPARED METHODS ON EXPERIMENTAL DATASETS

Datasets	AUC Values of (TPR, FPR)						
	DFAE	AAE	FrFE	PTA	LSDM -MOG	SSDF	OSP -GoDec
San Diego	**0.98594**	0.95928	0.98379	0.93919	0.96417	0.94537	0.97865
Texas Coast	**0.99790**	0.98901	0.98917	0.96419	0.99130	0.94854	0.98969
Gainesville	**0.99927**	0.99210	0.96626	0.93874	0.95177	0.88688	0.97116
Cat Island	**0.99828**	0.95361	0.97921	0.96886	0.97521	0.98104	0.97689
Average	**0.99535**	0.97350	0.97961	0.95275	0.97061	0.94046	0.97910
Datasets	AUC Values of (FPR, τ)						
	DFAE	AAE	FrFE	PTA	LSDM -MOG	SSDF	OSP -GoDec
San Diego	**0.00637**	0.02935	0.04428	0.11998	0.05322	0.09292	0.00800
Texas Coast	**0.00611**	0.01039	0.04351	0.16446	0.07825	0.12948	0.02361
Gainesville	**0.00272**	0.00840	0.03483	0.10156	0.10429	0.13545	0.02091
Cat Island	**0.00110**	0.00395	0.00767	0.02878	0.02606	0.09747	0.02735
Average	**0.00408**	0.01302	0.03257	0.10370	0.06546	0.11383	0.01997

TABLE V
EVALUATION AUC VALUES OF THE COMPARED METHODS ON THE LOS ANGELES DATASET

AUC Values	DFAE	AAE	FrFE	PTA	LSDM-MOG	SSDF	OSP-GoDec
(TPR, FPR)	**0.98721**	0.98031	0.97180	0.93002	0.95505	0.91362	0.95831
(FPR, τ)	**0.00368**	0.01013	0.02685	0.10378	0.07557	0.11382	0.04795

The lines at the top and bottom of each column are the maximum and minimum of the detection values, respectively, which are normalized. Thus, the distance between the position of the red box and blue box indicates anomaly–background separability. Compared with other methods, the distribution range of the background in the DFAE method is more concentrated, and there is always a gap between the two boxes on the majority of datasets. None of the four methods, FrFE, PTA, MOG, and OSP-GoDec has two completely separated boxes. In other words, the anomaly box always overlaps with the background box in these methods, which implies that misdetection or false alarm will occur in them. The AAE method can also distinguish anomalies and the background very well. But for the San Diego dataset, there is almost no gap between the anomaly and background boxes, which substantially affects the discriminability of anomalies. Therefore, it can conclude that the performance of the proposed DFAE method is more satisfactory in terms of separability of anomaly and background, which is consistent with the visual detection maps in Fig. 8 and the AUC values of (FPR, τ) in Table IV.

Besides, the computing time of each comparison method is recorded in seconds in Table VI. Since our proposed method

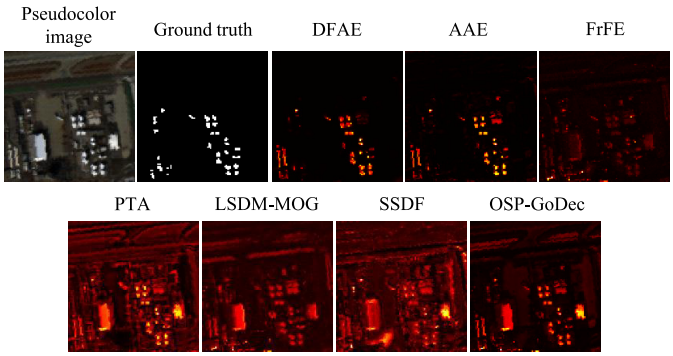


Fig. 11. Detection maps of the compared methods on the Los Angeles dataset.

TABLE VI
COMPUTING TIME (IN SECONDS) OF THE COMPARED METHOD

Datasets	DFAE	AAE	FrFE	PTA	LSDM -MOG	SSDF	OSP-GoDec
San Diego	10.98	308.67	29.20	44.66	**10.84**	1367.26	28.87
Texas Diego	**11.16**	371.41	22.46	46.45	33.29	1154.53	33.46
Gainesville	**10.54**	309.52	23.38	43.13	20.89	1358.61	31.02
Cat Island	**15.81**	655.13	56.99	80.89	54.44	1051.46	517.03
Average	**12.12**	411.18	33.01	53.78	29.87	1232.97	152.60

does not require retraining on different datasets, the running time of the DFAE method in Table VI is the time spent for the inference process, while the running time of the AAE method includes the time spent on training the HSI data, which may take more than 4 min. As shown in Table VI, our DFAE method is the most efficient among these seven HAD methods without considering the training duration. This finding again verifies the competitiveness of the proposed method in HAD.

To further evaluate the generalization ability of the DFAE method, we use another dataset, namely, Los Angeles, as a testing set with the fixed parameters in Section III-B. In other words, this dataset directly performs anomaly detection with the proposed method without participating in the process of parameter determination. The Los Angeles dataset was captured by the AVIRIS sensor with a spatial resolution of 7.1 m. It consists of 100 × 100 pixels and 205 spectral bands from 0.43 to 0.86 μm. Like other experimental datasets, its first 188 bands are selected for the experiments. The detection maps and the AUC values are illustrated in Fig. 11 and Table V, respectively. By comparing the detection results of different methods, we note that our DFAE method is highly

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competitive both in terms of visual effects and quantitative metrics. Besides, excellent results can also be achieved when other datasets are used as the training set, just adjusting some parameters. Nevertheless, under a given training set, all the testing sets still use the same set of parameters for anomaly detection. In essence, our proposed DFAE method has good generalization ability, which provides the possibility of deep-learning-based HAD in practical engineering applications.

IV. Conclusion

In this article, we develop a novel DFAE detection model for unsupervised HAD in the transformed HSI. The key point of our method is to detect anomalies in the frequency domain. To this end, several aspects have been taken into consideration. First, given the local similarity of HSI, a novel spectral rectification method is proposed to generate the LFCs, which can also alleviate the spectral variation problem. Besides, the Laplacian operator is used to extract the HFCs instead of directly subtracting the LFCs from the original image. Then, the proposed DFAE is learned to detect anomalies in the LFCs and the HFCs in parallel. Finally, anomalies of other hyperspectral datasets, including point and structural anomalies, are detected by the well-learned model with the parameters optimized by the trained HSI. The experimental results show that the proposed DFAE can not only offer great feasibility for real application but also achieve more satisfactory performance compared with the state-of-the-art methods in HAD.

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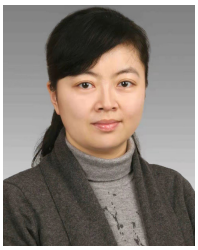
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